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Capacitance

- 25-1 Capacitors and Capacitance
- 25-2 Calculating the Capacitance
- 25-3 Capacitors in Series and Parallel
- 25-4 Energy Stored in An Electric-Field
- 25-5 Capacitor with A Dielectric



25-1 Capacitors and Capacitance

- A capacitor is a device that stores electrical charge and energy.
- During the charging process, a capacitor converts other forms of energy (i.e.: solar, chemical, mechanical energy, etc.) into electrical energy and stores it within the capacitor.
- Capacitors are widely used in circuits for buffering and tuning.



Paul Silvermann/Fundamental Photographs
Figure 25-1 An assortment of capacitors.

- **Two-conductor capacitor:** Composed of two conductors / electrodes separated by a certain medium or vacuum.

Symbol: 



25-1 Capacitors and Capacitance

Capacitor charging: The process of making two conductors carry equal amounts of opposite charges by connecting them to a power source.

Capacitance: $C = Q / V$ [F] (*farad*)

- ✿ The commonly used units of capacitors include μF , nF , and pF .
- ✿ The larger the value of **C**, the more charge and electrical energy the capacitor will store.
- ✿ The capacitance depends only on the shape, size, relative positions of the conductors, and dielectric of the capacitor, and is independent of **Q** and **V**.



25-2 Calculating the Capacitance

Problem solving strategy:

We generally determine the capacitance of a particular capacitor configuration by:

- (1) Assuming a charge q on the plates.
- (2) Finding the electric field $\vec{E}$ between the plates in terms of this charge, using Gauss' law.
- (3) Knowing $\vec{E}$, calculate the potential difference V between the plates.
- (4) Calculating C from $C = q/V$.



Example-1 A Parallel-Plate Capacitor

1. A parallel plate capacitor has two plates with area A , separated by a distance d that is small in comparison with their dimensions. Find the capacitance of this capacitor.

Solution:

$$C = Q / V$$

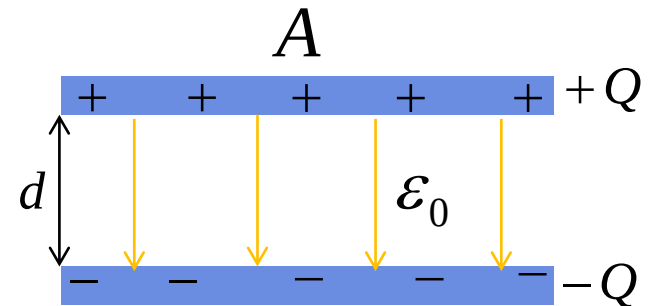
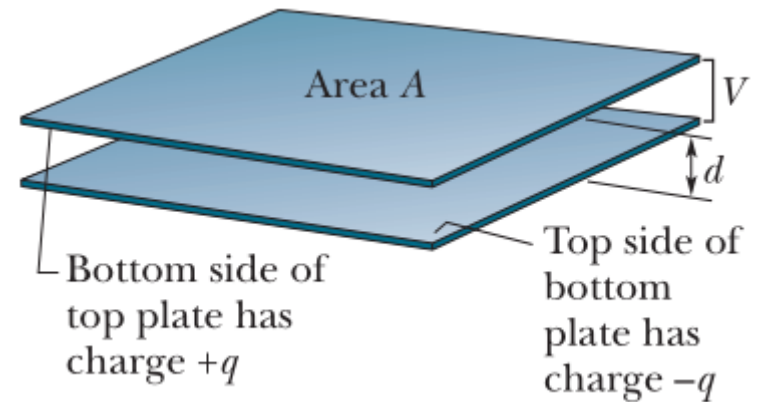
$$V = Ed$$

$$E = \sigma / \epsilon_0, \sigma = Q / A$$

$$E = Q / \epsilon_0 A$$

$$V = Qd / \epsilon_0 A$$

$$C = \epsilon_0 A / d$$



$$C = \epsilon_0 A / d$$



Example-2 A Spherical Capacitor

2. Two concentric spherical conducting shells are separated by vacuum. The inner shell has radius R_1 , and the outer shell has radius R_2 . Find the capacitance of this spherical capacitor.

Solution:

$$C = Q / V$$

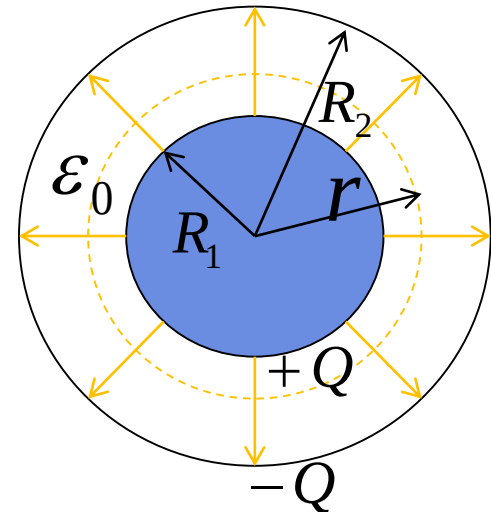
$$V = \int \vec{E} \cdot d\vec{l} = \int E dr$$

$$\oint \vec{E} \cdot d\vec{A} = E \cdot 4\pi r^2 = Q / \epsilon_0$$

$$E = Q / 4\pi\epsilon_0 r^2$$

$$V = \int_{R_1}^{R_2} E dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$C = \frac{4\pi\epsilon_0 R_1 R_2}{R_2 - R_1}$$



$$E = Q / 4\pi\epsilon_0 r^2$$



25-3 Capacitors in Series and Parallel

- Capacitors in Parallel

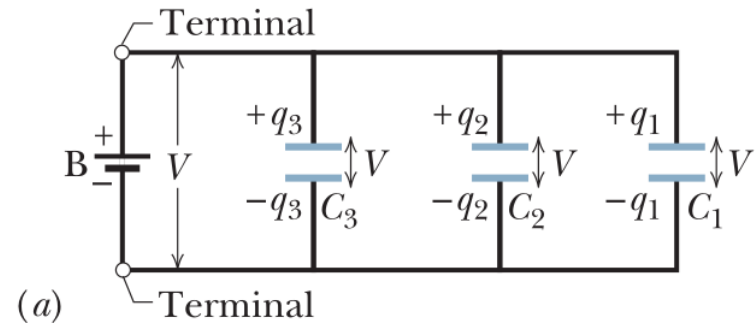
$$Q_1 + Q_2 + \dots = Q$$

$$V_1 = V_2 = \dots = V$$

Equivalent capacitance is :

$$C_{eq} = C_1 + C_2 + \dots$$

The parallel equivalent capacitance is greater than the capacitance of any of the capacitors connected in parallel.



Parallel capacitors and their equivalent have the same V ("par- V ").

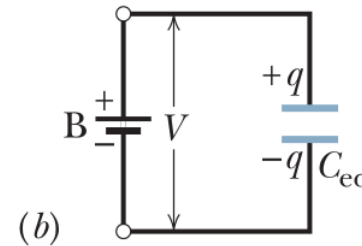


Figure 25-8 (a) Three capacitors connected in parallel to battery B. The battery maintains potential difference V across its terminals and thus across *each* capacitor. (b) The equivalent capacitor, with capacitance C_{eq} , replaces the parallel combination.



25-3 Capacitors in Series and Parallel

- Capacitors in Parallel

When a potential difference V is applied across several capacitors connected in parallel, that potential difference V is applied across each capacitor. The total charge q stored on the capacitors is the sum of the charges stored on all the capacitors.

To derive an expression for C_{eq} in Fig. 25-8b, we first use Eq. 25-1 to find the charge on each actual capacitor:

$$q_1 = C_1V, \quad q_2 = C_2V, \quad \text{and} \quad q_3 = C_3V.$$

The total charge on the parallel combination of Fig. 25-8a is then

$$q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V.$$

The equivalent capacitance, with the same total charge q and applied potential difference V as the combination, is then

$$C_{\text{eq}} = \frac{q}{V} = C_1 + C_2 + C_3,$$

a result that we can easily extend to any number n of capacitors, as

$$C_{\text{eq}} = \sum_{j=1}^n C_j \quad (n \text{ capacitors in parallel}). \quad (25-19)$$



25-3 Capacitors in Series and Parallel

- Capacitors in Series

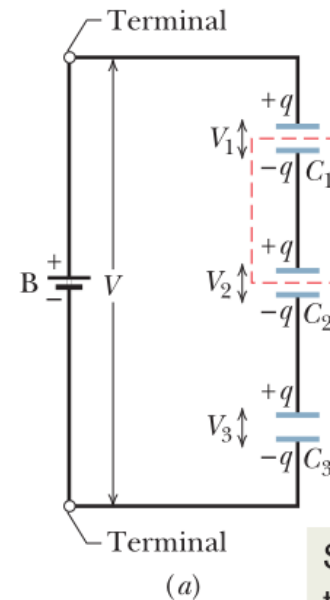
$$Q_1 = Q_2 = \dots = Q$$

$$V_1 + V_2 + \dots = V$$

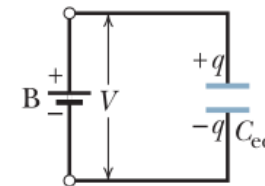
Equivalent capacitance is:

$$1/C_{eq} = 1/C_1 + 1/C_2 + \dots$$

The series equivalent capacitance is less than the capacitance of any of the capacitors connected in series.



(a)



(b)

Series capacitors and their equivalent have the same q ("seri-q").

Figure 25-9 (a) Three capacitors connected in series to battery B. The battery maintains potential difference V between the top and bottom plates of the series combination. (b) The equivalent capacitor, with capacitance C_{eq} , replaces the series combination.



25-3 Capacitors in Series and Parallel

• Capacitors in Series



When a potential difference V is applied across several capacitors connected in series, the capacitors have identical charge q . The sum of the potential differences across all the capacitors is equal to the applied potential difference V .

To derive an expression for C_{eq} in Fig. 25-9b, we first use Eq. 25-1 to find the potential difference of each actual capacitor:

$$V_1 = \frac{q}{C_1}, \quad V_2 = \frac{q}{C_2}, \quad \text{and} \quad V_3 = \frac{q}{C_3}.$$

The total potential difference V due to the battery is the sum

$$V = V_1 + V_2 + V_3 = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right).$$

The equivalent capacitance is then

$$C_{\text{eq}} = \frac{q}{V} = \frac{1}{1/C_1 + 1/C_2 + 1/C_3},$$

or

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}.$$

We can easily extend this to any number n of capacitors as

$$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^n \frac{1}{C_j} \quad (n \text{ capacitors in series}). \quad (25-20)$$



25-4 Energy Stored in An Electric-Field

A capacitor is an important energy storage device. During the charging process (i.e., the establishment of a charge system), external forces do work against the electric field force, converting other forms of energy into electrical energy and storing it in the electric field inside the capacitor.

**Energy Storage
in Capacitors :**

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$$

★ According to the equation, $Q=CV$, $U=CV^2/2$, it can be seen that capacitance **reflects the capacitor's ability to store charge and energy.**

★ Energy density of electric field is $u = \frac{1}{2} \varepsilon_0 E^2$

★ The energy of a capacitor is stored in the electric field, it is the energy of the electric field.



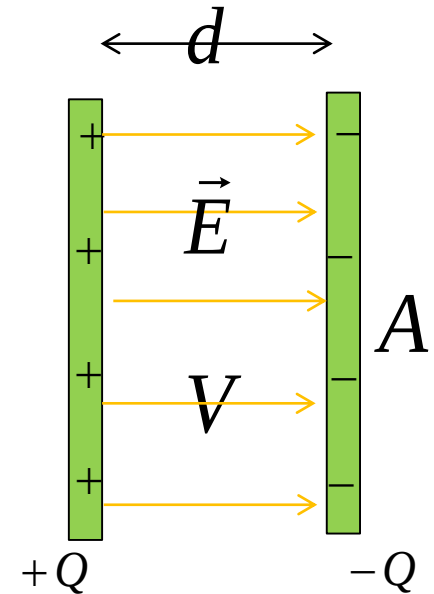
25-4 Energy Stored in An Electric-Field

Energy density of electric field:

$$V = Ed, \quad E = \sigma / \epsilon_0$$

$$U = \frac{1}{2} QV = \frac{1}{2} (\sigma A)(Ed) = \frac{1}{2} \epsilon_0 E^2 Ad$$

$$u = U / Ad = \frac{1}{2} \epsilon_0 E^2$$

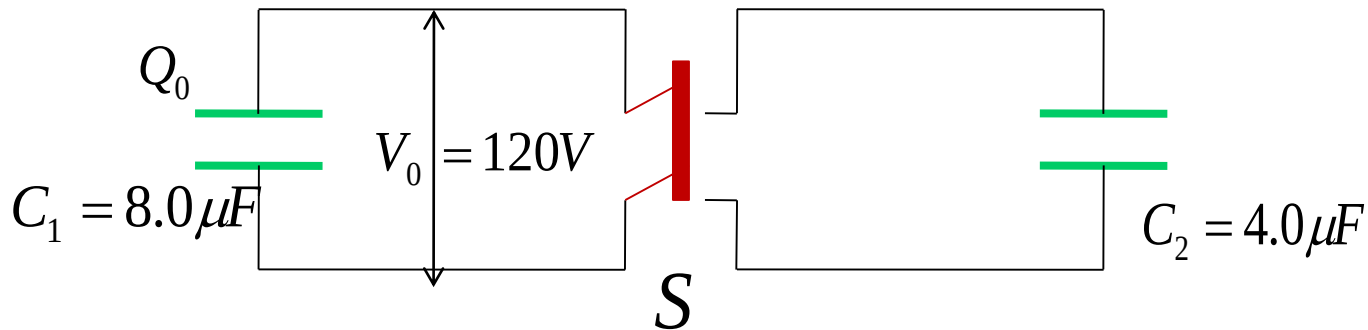


$$u = \frac{1}{2} \epsilon_0 E^2$$



Example-4

4. We charge a capacitor of capacitance $C_1 = 8.0\mu F$ by connecting it to a source of potential difference $V_0 = 120V$. The switch S is initially open. Once C_1 is charged, the source of potential difference is disconnected.
- (a) what is the charge Q_0 on C_1 if switch S is left open?
 - (b) what is the energy stored in C_1 if switch S is left open?
 - (c) The capacitor of capacitance $C_2 = 4.0\mu F$ is initially uncharged. After we close switch S , what is the potential difference across each capacitor, and what is the charge on each capacitor?
 - (d) what is the total energy of the system after we close switch S .

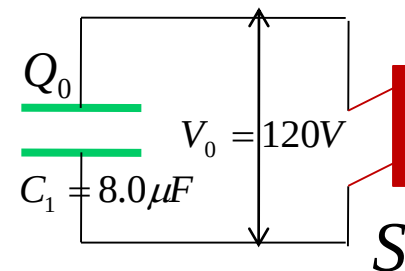


Example-4

Solution:

$$a) Q_0 = C_1 V_0 = 8.0 \times 120 = 960 \mu C$$

$$b) U_0 = \frac{1}{2} Q_0 V_0 = \frac{1}{2} \times 960 \times 10^{-6} \times 120 = 0.058 J$$

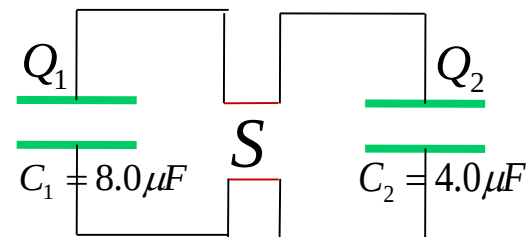


$$c) Q_1 + Q_2 = Q_0, \quad Q_1 / Q_2 = C_1 / C_2 = 2.0$$

$$Q_1 = \frac{2}{3} Q_0 = 640 \mu C, \quad V_1 = Q_1 / C_1 = 80V$$

$$Q_2 = \frac{1}{3} Q_0 = 320 \mu C, \quad V_2 = V_1 = V = 80V$$

$$d) U = \frac{1}{2} Q_1 V + \frac{1}{2} Q_2 V = \frac{1}{2} Q_0 V = 0.038 J$$

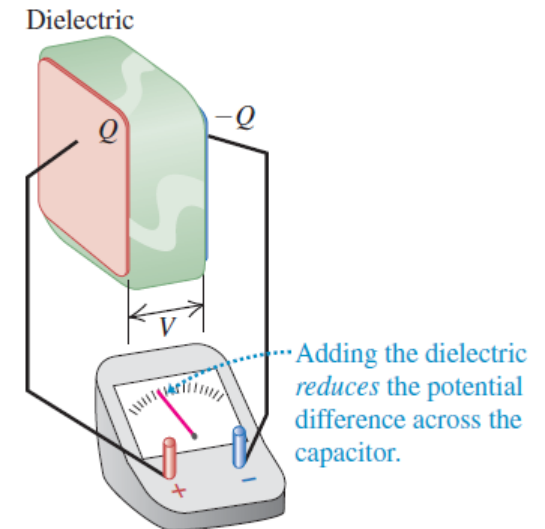
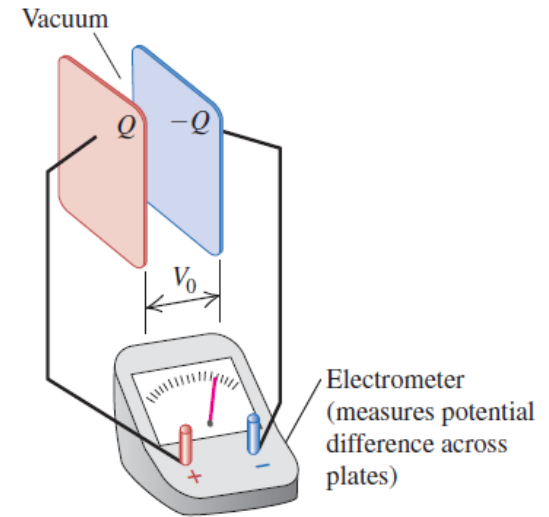


25-5 Capacitor with A Dielectric

Dielectrics are insulating materials.

Most capacitors are filled with a dielectric material, and **the functions of the dielectric material** are as follow:

1. By bringing the two conductors of a capacitor very close together without them touching, and increase the capacitance of the capacitor.
2. Another effect of the introduction of a dielectric is to limit the potential difference that can be applied between the plates to a certain value V_{max} , called the breakdown potential.



$$C = \epsilon_0 A / d, C = Q / V$$



25-5 Capacitor with A Dielectric

tance? Michael Faraday—to whom the whole concept of capacitance is largely due and for whom the SI unit of capacitance is named—first looked into this matter in 1837. Using simple equipment much like that shown in Fig. 25-12, he found that the capacitance *increased* by a numerical factor κ , which he called the **dielectric constant** of the insulating material.



The Royal Institute, England/Bridgeman Art Library/NY

Figure 25-12 The simple electrostatic apparatus used by Faraday. An assembled apparatus (second from left) forms a spherical capacitor consisting of a central brass ball and a concentric brass shell. Faraday placed dielectric materials in the space between the ball and the shell.



25-5 Capacitor with A Dielectric

Dielectric constant: Capacitance of the capacitors containing dielectric

$$K = C / C_0$$

Capacitance of capacitor in vacuum

Permittivity $\epsilon = K \epsilon_0$

Dielectric constant $K = \epsilon / \epsilon_0$

Table 24.1 Values of Dielectric Constant K at 20°C

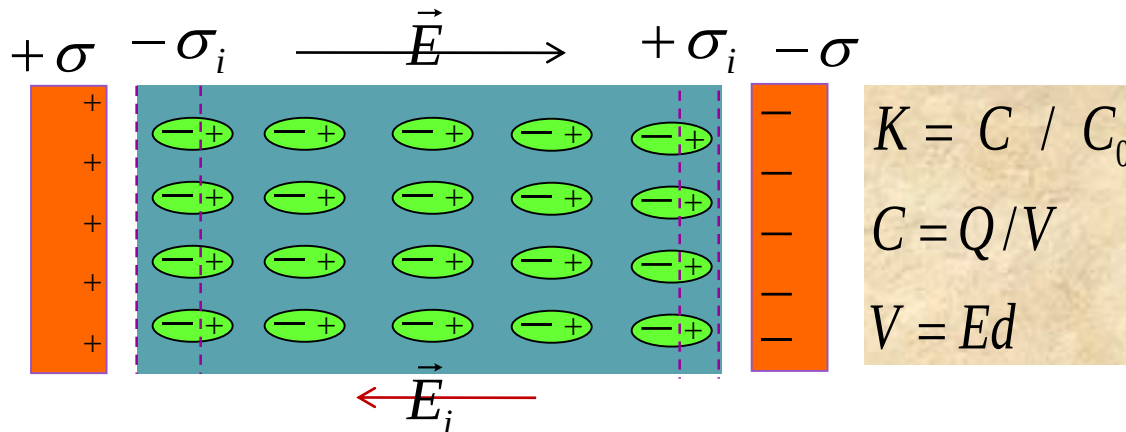
Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310



25-5 Capacitor with A Dielectric

Polarization of Dielectric

When bound charges inside a dielectric are redistributed under the influence of an electric field, a net bound charge is generated inside and on the surface of the dielectric. This charge is called polarization charge (an induced charge), and this phenomenon is called dielectric polarization.



The electric field generated by the polarization charge inside the medium is opposite in direction to the original electric field, thus canceling out part of the electric field.

$$\sigma_i = -\sigma\left(1 - \frac{1}{K}\right), \quad Q_i = -Q\left(1 - \frac{1}{K}\right)$$



25-5 Capacitor with A Dielectric

- No dielectric material is absolutely non-conductive, capacitors always have leakage current during usage.
- **Dielectric breakdown:** under a sufficiently strong electric field, electrons are pulled out and collide with other molecules, releasing even more electrons. This instantaneous avalanche of electrons creates a spark/arc discharge, causing a short circuit phenomenon.

Table 24.2 Dielectric Constant and Dielectric Strength of Some Insulating Materials

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex glass	4.7	1×10^7



Dielectric breakdown



Current and Resistance

- 26-1 Electric current
- 26-2 Current density
- 26-3 Resistivity (Ohm's Law)
- 26-4 Resistance
- 26-5 Electromotive Force and Circuit
- 26-6 Energy and Power in Electric Circuits



26-1 Electric Current

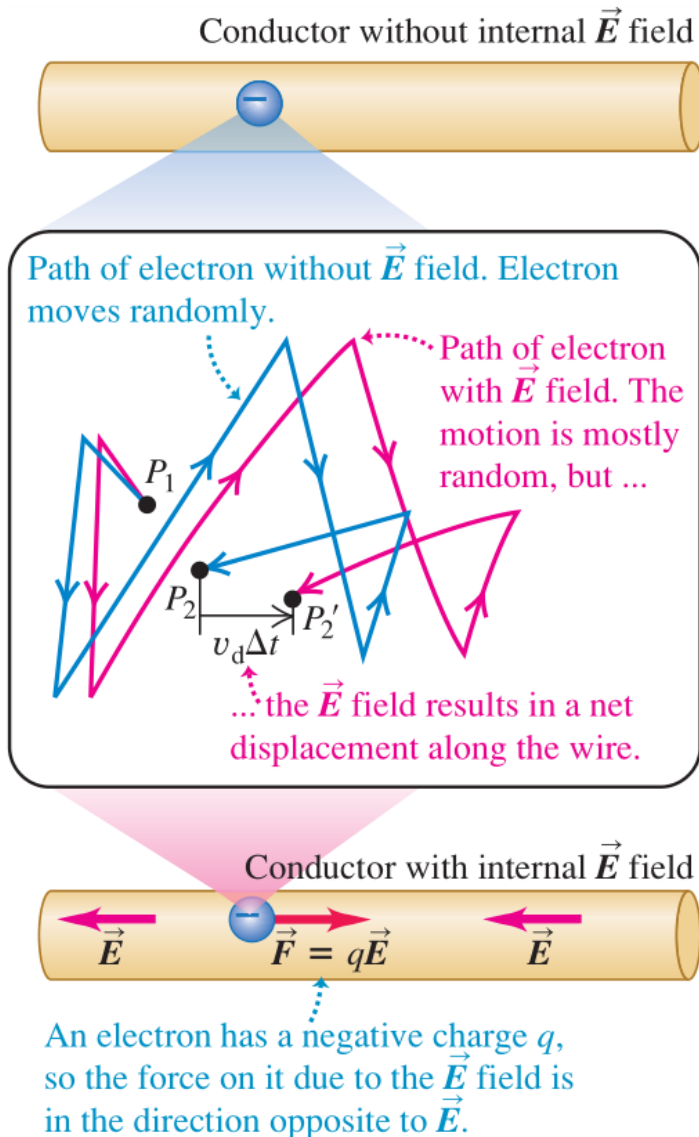
Current: The directional movement of electric charges forms an electric current.

In electrostatic situations, the electric field is zero everywhere within the conductor, and there is no current. However, this does not mean that all charges within the conductor are at rest.

In an ordinary metal such as copper or aluminum, some of the electrons are free to move within the conducting material. These free electrons move randomly in all directions, somewhat like the molecules of a gas but with much greater speeds, of the order 10^6 m/s. The electrons nonetheless do not escape from the conducting material, because they are attracted to the positive ions of the material. The motion of the electrons is random, so there is *no net flow* of charge in any direction and hence no current.



26-1 Electric Current



25.1 If there is no electric field inside a conductor, an electron moves randomly from point P_1 to point P_2 in a time Δt . If an electric field $\vec{E}$ is present, the electric force $\vec{F} = q\vec{E}$ imposes a small drift (greatly exaggerated here) that takes the electron to point P'_2 , a distance $v_d \Delta t$ from P_2 in the direction of the force.

When an external electric field $\vec{E}$ is applied, electrons undergo collective superposition and directional drift motion (order of 10^{-4} m/s) based on their high-speed thermal motion, thus forming an electric current. These electrons are called **current-carrying charges**.



26-1 Electric Current

- Different conductors have different charge carriers. The charge carriers of ionized gases and electrolytes are positive and negative ions, while the charge carriers of semiconductors are electrons and holes.
- While the random motion of the electrons has a very fast average speed of about 10^6 m/s, the drift speed is very slow, often on the order of 10^{-4} m/s.

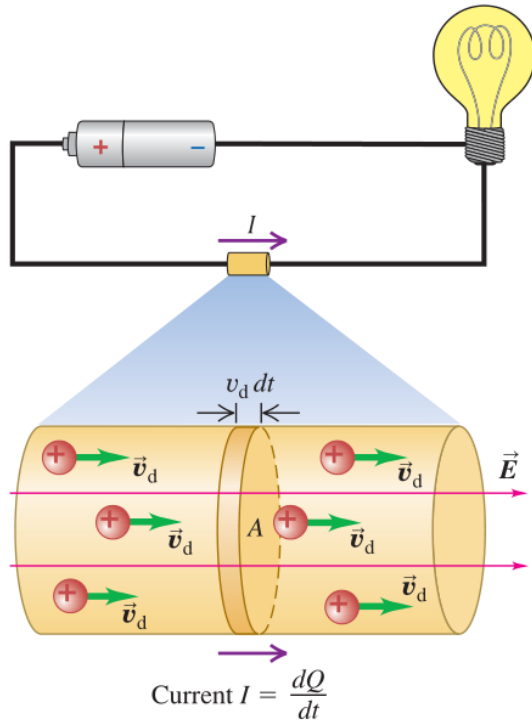
Question:

Given that the electrons move so slowly, why the light comes on immediately when you turn on the switch of a flashlight?

- The reason is that the **electric field** is set up in the wire with a speed approaching the speed of light, and electrons start to move all along the wire at very nearly the same time.



26-2 Current and Current density



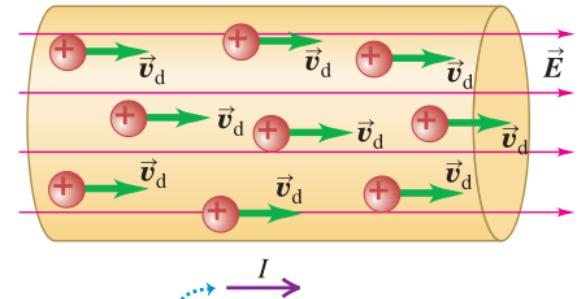
Current:

$$I = dQ / dt \quad [\text{A}]$$

SI unit of current:
1 A = 1 C/s

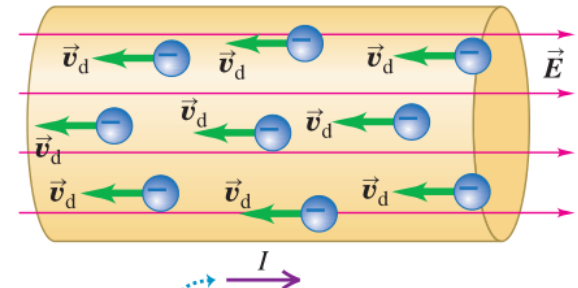
The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive or negative.

(a)



A **conventional current** is treated as a flow of positive charges, regardless of whether the free charges in the conductor are positive, negative, or both.

(b)



In a metallic conductor, the moving charges are electrons — but the *current* still points in the direction positive charges would flow.



26-2 Current and Current density

Current: $I = dQ / dt$ [A]

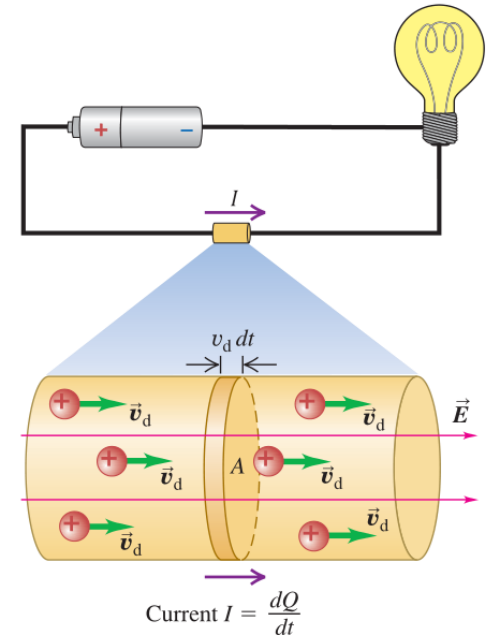
✪ **Magnitude:** Current I is the time rate of charge transfer through the cross-sectional area A .

✪ **Direction:** The direction of the directional movement of positive charges.

Current density: $\vec{J}$

✪ **Magnitude:** Current intensity per unit cross-sectional area at this point

✪ **Direction:** The direction of the directional movement of the positive charge at this point.

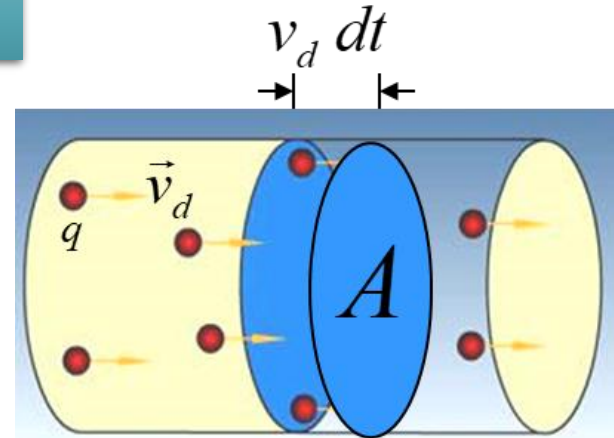


26-2 Current and Current density

Current density vs. drift velocity:

$$\vec{J} = nq\vec{v}_d$$


Carrier concentration / density



Suppose there are n moving charged particles per unit volume. We call n the **concentration** of particles; its SI unit is m^{-3} . Assume that all the particles move with the same drift velocity with magnitude v_d . In a time interval dt , each particle moves a distance $v_d dt$. The particles that flow out of the right end of the shaded cylinder with length $v_d dt$ during dt are the particles that were within this cylinder at the beginning of the interval dt . The volume of the cylinder is $Av_d dt$, and the number of particles within it is $nAv_d dt$. If each particle has a charge q , the charge dQ that flows out of the end of the cylinder during time dt is

$$dQ = q(nAv_d dt) = nqv_d A dt$$



26-2 Current and Current density

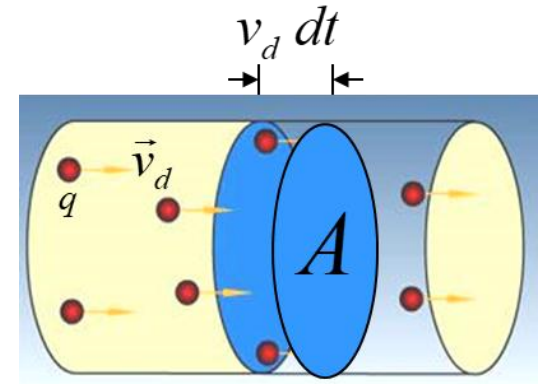
and the current is

$$I = \frac{dQ}{dt} = nqv_d A$$

The current *per unit cross-sectional area* is called the **current density** J :

$$J = \frac{I}{A} = nqv_d$$

The units of current density are amperes per square meter (A/m^2).



We can also define a *vector* current density $\vec{J}$ that includes the direction of the drift velocity:

$$\vec{J} = nq\vec{v}_d \quad (\text{vector current density}) \quad (25.4)$$

There are *no* absolute value signs in Eq. (25.4). If q is positive, $\vec{v}_d$ is in the same direction as $\vec{E}$; if q is negative, $\vec{v}_d$ is opposite to $\vec{E}$. In either case, $\vec{J}$ is in the same direction as $\vec{E}$. Equation (25.3) gives the *magnitude* J of the vector current density $\vec{J}$.



26-2 Current and Current density

Current density vs. current

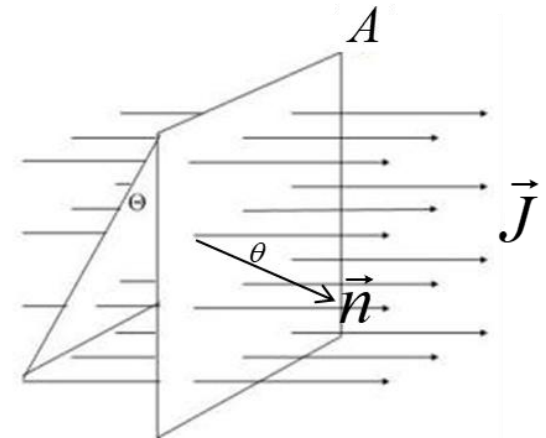
$$I = \int \vec{J} \cdot d\vec{A}$$

Relationship between vector and its flux.

Situation 1 . If $\vec{J}$ is uniformly distributed and passes through plane surface

$$I = JA$$

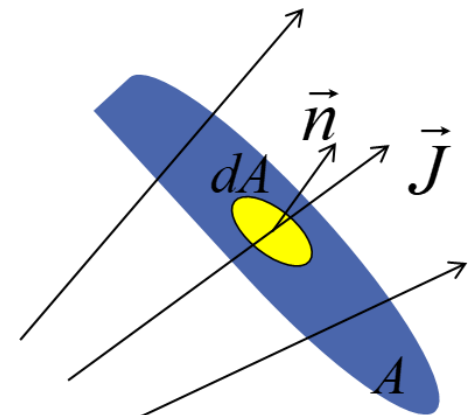
$$I = JA \cos \theta = \vec{J} \cdot \vec{A}$$



Situation 2 . If $\vec{J}$ is not uniformly distributed and passes through curved surface

$$dI = \vec{J} \cdot d\vec{A}$$

$$I = \int \vec{J} \cdot d\vec{A}$$



Example-1

1. An 18 gauge copper wire has a nominal diameter of 1.02mm. This wire carries a constant current of 1.67A to a 200-watt lamp. The density of free electrons is 8.5×10^{28} electrons per cubic meter. Find the magnitudes of (a) the current density and (b) the drift velocity.

Solution:

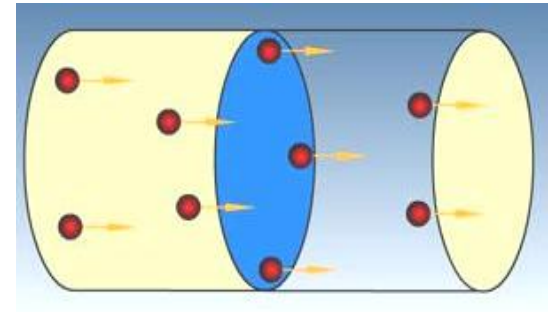
a) $J = I / A$

$$A = \pi r^2 = 3.14 \times (1.02 / 2)^2 = 8.17 \times 10^{-7} \text{ m}^2$$

$$J = \frac{I}{A} = \frac{1.67}{8.17 \times 10^{-7}} = 2.04 \times 10^6 \text{ A/m}^2$$

b) $J = nq\vec{v}_d$

$$\begin{aligned} v_d &= J / nq = \frac{2.04 \times 10^6}{8.5 \times 10^{28} \times 1.60 \times 10^{-19}} \\ &= 1.5 \times 10^{-4} \text{ m/s} = 0.15 \text{ mm/s} \end{aligned}$$



$$\vec{J} = nq\vec{v}_d$$

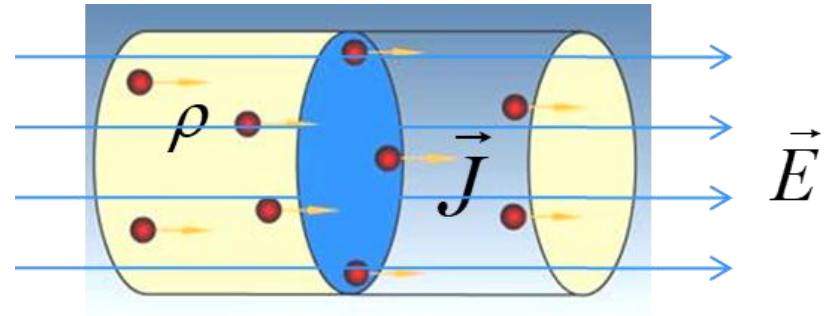


26-3 Resistivity

Ohm's Law

(in Differential form):

$$\vec{E} = \rho \vec{J}$$



- ✦ Ohm's law holds approximately true only for linear materials under certain conditions.
- ✦ ρ is the resistivity of materials, SI unit is $[\Omega \cdot \text{m}]$
- ✦ The reciprocal of resistivity is conductivity: $\sigma = \frac{1}{\rho}$
(Good electrical conductive materials are usually good thermal conductive materials)
- ✦ Resistivity changes with temperature.

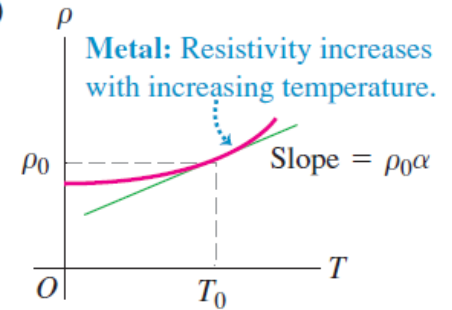


26-3 Resistivity

☀ Resistivity changes with temperature. (a)

$$\rho(T) = \rho_0 [1 + \alpha(T - T_0)] \quad (\Delta T < 100 \text{ C}^\circ)$$

α *temperature coefficient of resistivity*

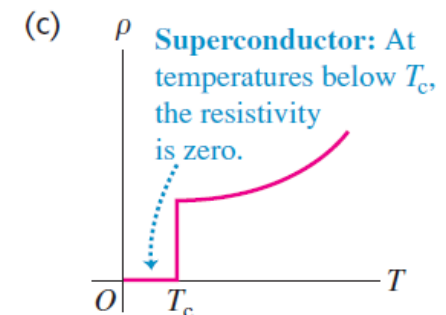
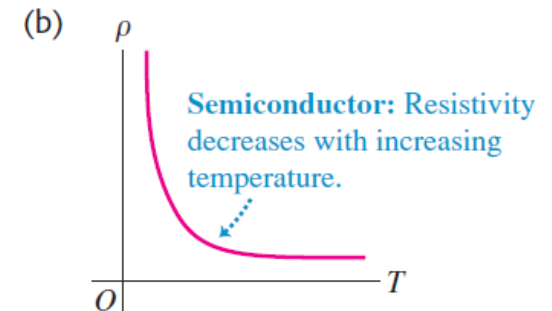


For metal:

$\alpha > 0$ Resistivity increases with increasing temperature.

For semiconductor:

$\alpha < 0$ Resistivity decreases as temperature increases.



26-3 Resistivity

Table 26-1 Resistivities of Some Materials at Room Temperature (20°C)

Material	Resistivity, ρ ($\Omega \cdot \text{m}$)	Temperature Coefficient of Resistivity, α (K^{-1})
<i>Typical Metals</i>		
Silver	1.62×10^{-8}	4.1×10^{-3}
Copper	1.69×10^{-8}	4.3×10^{-3}
Gold	2.35×10^{-8}	4.0×10^{-3}
Aluminum	2.75×10^{-8}	4.4×10^{-3}
Manganin ^a	4.82×10^{-8}	0.002×10^{-3}
Tungsten	5.25×10^{-8}	4.5×10^{-3}
Iron	9.68×10^{-8}	6.5×10^{-3}
Platinum	10.6×10^{-8}	3.9×10^{-3}
<i>Typical Semiconductors</i>		
Silicon, pure	2.5×10^3	-70×10^{-3}
Silicon, <i>n</i> -type ^b	8.7×10^{-4}	
Silicon, <i>p</i> -type ^c	2.8×10^{-3}	
<i>Typical Insulators</i>		
Glass	$10^{10} - 10^{14}$	
Fused quartz	$\sim 10^{16}$	

We can write Eq. 26-10 in vector form as

$$\vec{E} = \rho \vec{J}. \quad (26-11)$$

Equations 26-10 and 26-11 hold only for *isotropic* materials—materials whose electrical properties are the same in all directions.

We often speak of the **conductivity** σ of a material. This is simply the reciprocal of its resistivity, so

$$\sigma = \frac{1}{\rho} \quad (\text{definition of } \sigma). \quad (26-12)$$

The SI unit of conductivity is the reciprocal ohm-meter, $(\Omega \cdot \text{m})^{-1}$. The unit name mhos per meter is sometimes used (mho is ohm backwards). The definition of σ allows us to write Eq. 26-11 in the alternative form

$$\vec{J} = \sigma \vec{E}. \quad (26-13)$$



26-4 Resistance

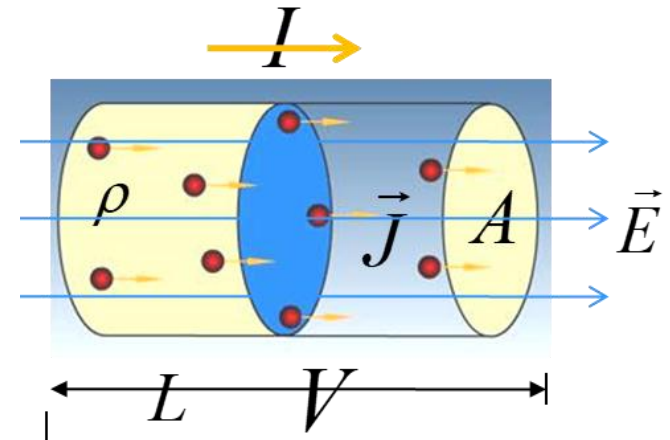
Ohm's Law:

(Integral form)

$$V = RI$$

Resistance:

$$R = \int \frac{\rho}{A} dl \quad [\Omega]$$



- ✦ Resistance of a conductor with a uniform cross-section: $R = \rho L / A$
- ✦ Resistance changes with temperature: $R(T) = R_0 [1 + \alpha(T - T_0)]$
- ✦ The voltage and current of a linear element are directly proportional.



Example-2

2. Suppose the resistance of a copper wire is $1.05\ \Omega$ at a temperature of 20°C . Find the resistance at 0°C and at 100°C .

Solution:

$$\alpha = 0.00393\ (\text{C}^\circ)^{-1}$$

$$R(T) = R_0[1 + \alpha(T - T_0)]$$

$$R(0^\circ\text{C}) = 1.05 \times [1 + 0.00393 \times (0 - 20)] = 0.97\ \Omega$$

$$R(100^\circ\text{C}) = 1.05 \times [1 + 0.00393 \times (100 - 20)] = 1.38\ \Omega$$

**Table 25.2 Temperature Coefficients of Resistivity
(Approximate Values Near Room Temperature)**

Material	$\alpha\ [(\text{C}^\circ)^{-1}]$	Material	$\alpha\ [(\text{C}^\circ)^{-1}]$
Aluminum	0.0039	Lead	0.0043
Brass	0.0020	Manganin	0.00000
Carbon (graphite)	-0.0005	Mercury	0.00088
Constantan	0.00001	Nichrome	0.0004
Copper	0.00393	Silver	0.0038
Iron	0.0050	Tungsten	0.0045



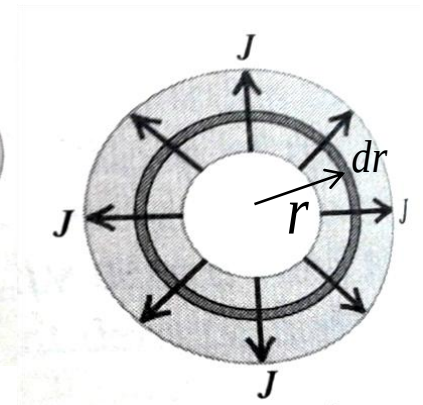
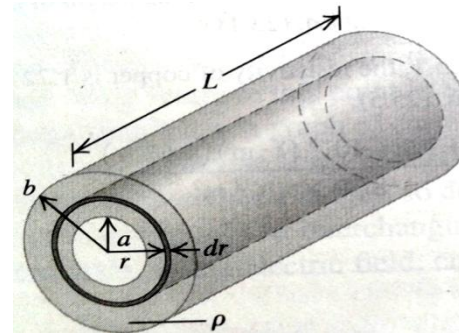
Example-3

3. The hollow cylinder has length L and inner and outer radius a and b . It is made of a material with resistivity ρ . A potential difference is set up between the inner and outer surfaces of the cylinder so that current flows radially through the cylinder. What is the resistance to this radial current flow?

Solution:

$$R = \int \frac{\rho}{A} dl$$

$$R = \int_a^b \frac{\rho}{2\pi r L} dr = \frac{\rho}{2\pi L} \ln \frac{b}{a}$$

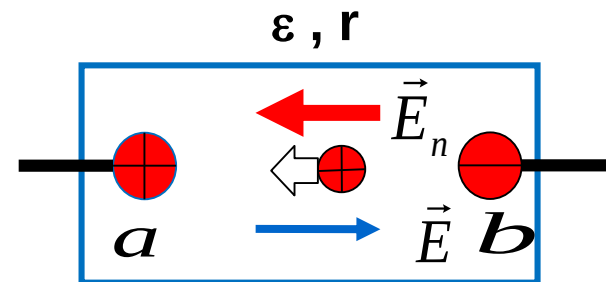
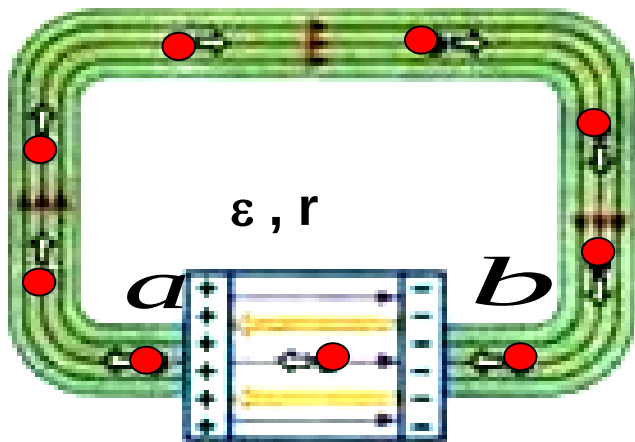


26-5 Electromotive Force and Circuit

A power source is required for the continuous flow of electric current. A power source is a device that has the ability to drive the positive charges from a low potential to a high potential—electromotive force (*emf.*), and convert other forms of energy into electrical energy. i.e.: batteries, electric generators, solar cells

Electromotive force (ε): ε is equal to the work done by the non-electrostatic force that moves a unit positive charge from the negative terminal to the positive terminal inside the power source.

Internal resistance (r): The internal resistance of the power supply changes with temperature and usage time.



$$\varepsilon = \int_{-}^{+} \vec{E}_n \cdot d\vec{l}$$



26-5 Electromotive Force and Circuit

Close-circuit: $V_{ab} = \mathcal{E} - Ir$

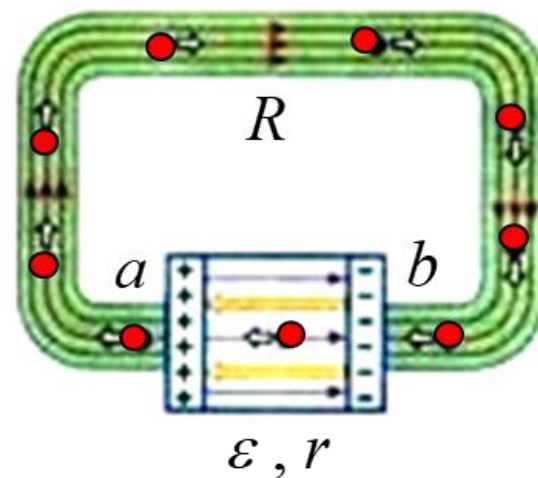
$$V_{ab} = IR$$

Open-circuit: $\mathcal{E} = V_{ab}$

Ohm's Law for the entire circuit: :

$$\mathcal{E} = I(R + r)$$

(V_{ab} is the power supply voltage, R is the external resistance)



$$\begin{aligned} \vec{E} + \vec{E}_n &= \rho \vec{J} \\ \oint (\vec{E} + \vec{E}_n) \cdot d\vec{l} &= \int_{-}^{+} \vec{E}_n \cdot d\vec{l} \\ \mathcal{E} &= \oint \rho \vec{J} \cdot d\vec{l} = \oint \rho J dl \\ &= \oint \frac{\rho J A}{A} dl = I \oint \frac{\rho}{A} dl \\ &= I(R + r) \end{aligned}$$



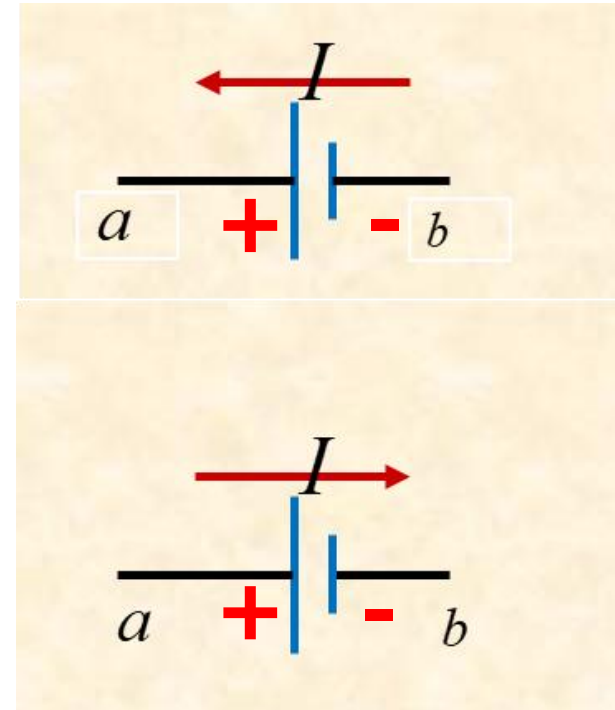
26-6 Energy and Power in Electric Circuits

- **Output power (discharge):**

$$P = V_{ab} I = \mathcal{E} I - I^2 r$$

- **Input power (being charged):**

$$P = V_{ab} I = \mathcal{E} I + I^2 r$$



From the direction of current I :

Low potential $\rightarrow$ High potential (Discharge)

High potential $\rightarrow$ Low potential (Being charged)



C H A P T E R 2 7

Circuits

- 27-1 Resistors in Series and Parallel
- 27-2 Kirchhoff's Rules
- 27-3 Electrical Measuring Instruments
- 27-4 RC Circuits



27-1 Resistors in Series and Parallel

Resistors in Series:

$$I_1 = I_2 = \dots = I \quad , \quad V_1 + V_2 + \dots = V$$

$$R_{eq} = R_1 + R_2 + \dots$$

Resistors in parallel:

$$I_1 + I_2 + \dots = I \quad , \quad V_1 = V_2 = \dots = V$$

$$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$$



27-2 Kirchhoff's Rules

Some resistive networks are not simply connected in series and parallel, requiring specific methods for analysis and calculation. Let's introduce the analytical and calculation methods of German physicist *Gustav Robert Kirchhoff*.

- **Junction:**

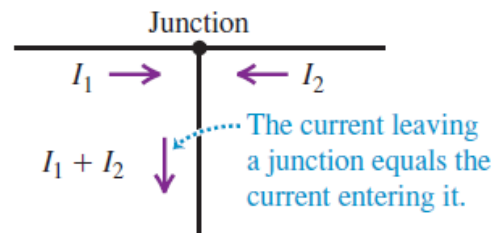
Connection point of three or more wires.

- **Loop:** Any closed circuit in a circuit.

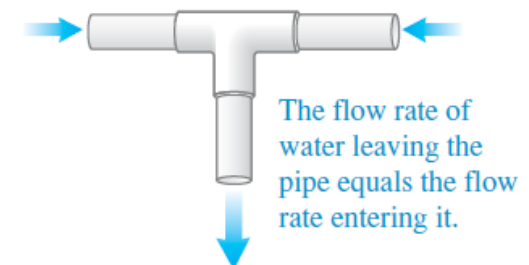
- **Sign conventions:**

1. The current flowing into a junction is positive, and the current flowing out is negative.
2. The direction in which the circuit circulates is firstly defined: along this direction, voltage rise is positive, and voltage drop is negative.

(a) Kirchhoff's junction rule



(b) Water-pipe analogy



(a) Sign conventions for emfs

$+\mathcal{E}$: Travel direction from $-$ to $+$:

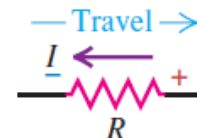


$-\mathcal{E}$: Travel direction from $+$ to $-$:

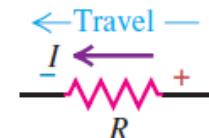


(b) Sign conventions for resistors

$+IR$: Travel *opposite* to current direction:



$-IR$: Travel *in* current direction:

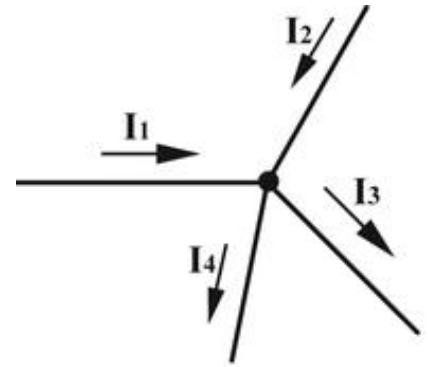


27-2 Kirchhoff's Rules

- Kirchhoff's junction rule**
(Conservation of Charge):

The algebraic sum of all currents intensities at the input and output junction is zero.

$$\sum I_i = 0$$

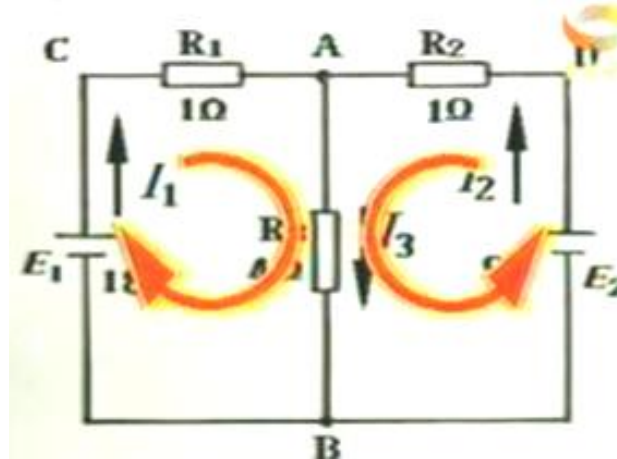


JUNCTION RULE: The sum of the currents entering any junction must be equal to the sum of the currents leaving that junction.

- Kirchhoff's loop rule**
(Conservation of Energy):

The algebraic sum of all voltages along any loop is zero.

$$\sum V_i = 0$$



27-2 Kirchhoff's Rules

Problem-Solving Strategy – Kirchhoff's Rules

- Draw the circuit diagram and assign labels and symbols to all known and unknown quantities
- Assign directions to the currents.
- Apply the junction rule to any junction in the circuit
- Apply the loop rule to as many loops as are needed to solve for the unknowns
- Solve the equations simultaneously for the unknown quantities
- Check your answers



Example-1

1. A bridge circuit as shown in the figure below. Find the current in each resistor and the equivalent resistance of the network of five resistors.

Solution:

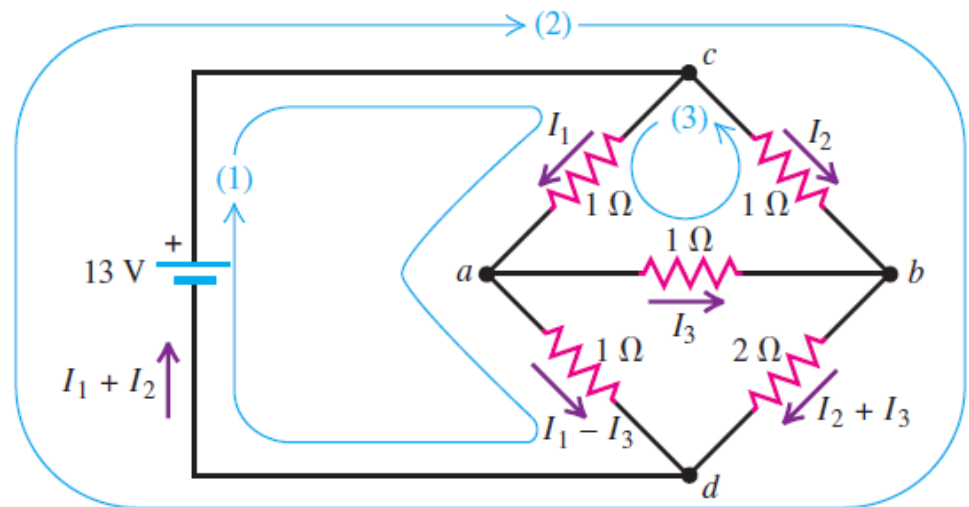
$$\begin{cases} 13 - I_1 \cdot 1 - (I_1 - I_3) \cdot 1 = 0 \\ 13 - I_2 \cdot 1 - (I_2 + I_3) \cdot 2 = 0 \\ -I_1 \cdot 1 - I_3 \cdot 1 + I_2 \cdot 1 = 0 \end{cases}$$

$$I_1 = 6A, I_2 = 5A, I_3 = -1A$$

$$I_1 - I_3 = 7A, I_2 + I_3 = 4A$$

$$R_{eq} = \varepsilon / (I_1 + I_2) = 13 / 11 = 1.2\Omega$$

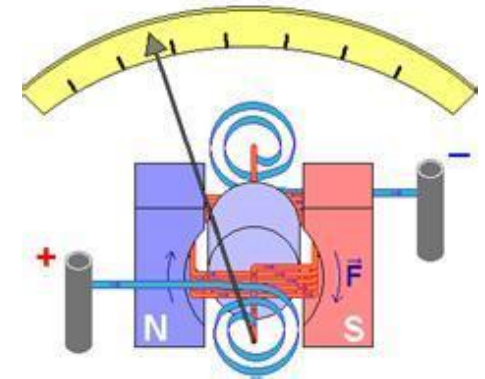
26.12 A network circuit with several resistors.



27-3 Electrical Measuring Instruments

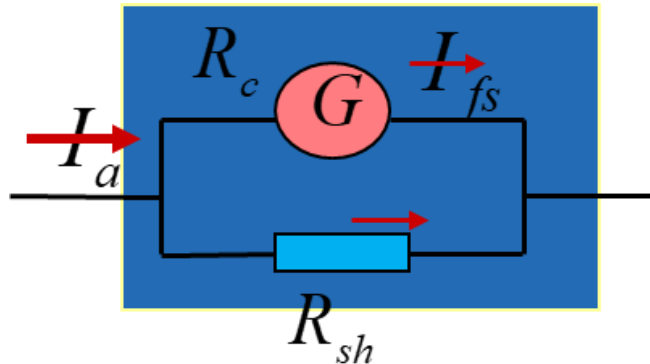
Some commonly used instruments, such as ammeters, voltmeters, ohmmeters, and automotive instruments, all use *galvanometers* (d'Arsonval galvanometer), or *meters*.

- Current for full-scale deflection: I_{fs}
- Resistance of the coil: R_c



Ammeters:

Connect the *galvanometer* in parallel with a *shunt resistor* (R_{sh}) to form an ammeter for measuring current. When the meter pointer deflects to full scale, it corresponds to the actual maximum range I_a .



$$I_{fs} R_c = (I_a - I_{fs}) R_{sh}$$



Example-2

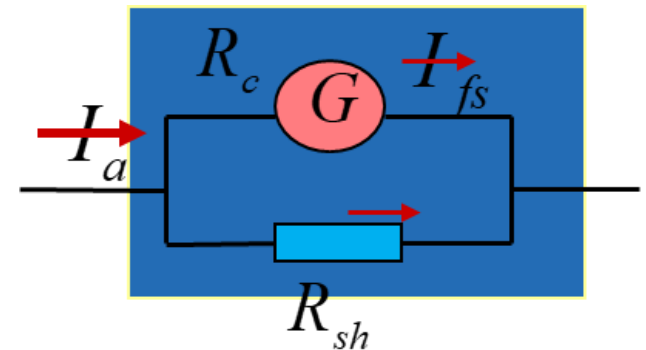
3. What shunt resistance is required to make the 1.00mA, 20.0Ω meter described above into an ammeter with a range of 0 to 50.0 mA.

Solution:

$$I_{fs} R_c = (I_a - I_{fs}) R_{sh}$$

$$1.00 \times 20.0 = (50.0 - 1.00) R_{sh}$$

$$R_{sh} = 20.0 / 49.0 = 0.408\Omega$$



Therefore, when an ammeter is connected to a circuit, its resistance value is often ignored, which is equivalent to a short circuit.

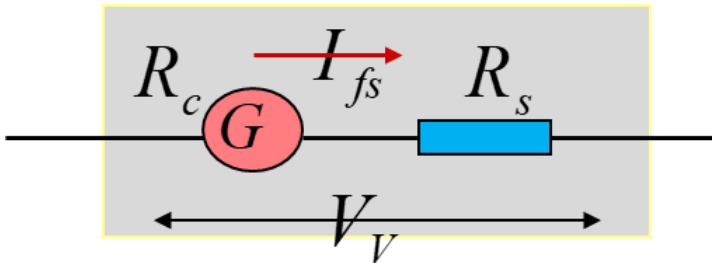
$$I_{fs} R_c = (I_a - I_{fs}) R_{sh}$$



27-3 Electrical Measuring Instruments

Voltmeters:

A voltmeter is constructed by connecting the *galvanometer* in series with voltage divider resistors R_s . When the pointer of the *galvanometer* deflects to full scale, it corresponds to the actual maximum range V_V .



$$V_V = I_{fs} (R_c + R_s)$$



Example-3

How can we make a 1.00mA, 20.0Ω galvanometer into a voltmeter with a maximum range of 10.0 V.

Solution:

$$V_V = I_{fs} (R_c + R_s)$$

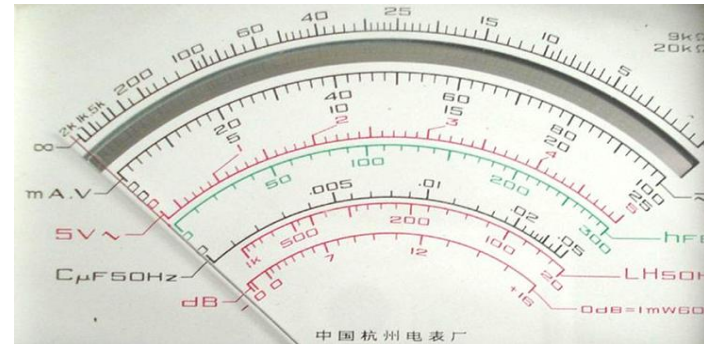
$$10.0 = 1.00 \times 10^{-3} \times (20.0 + R_s)$$

$$R_s = 10000 - 20.0 = 9980\Omega$$

1000 ohms-per-volt meter



Ohmmeters



- ☀ The reading direction of an ohmmeter is opposite to that of an Ammeter or Voltmeter.
- ☀ More accurate instruments are digital, meaning the meter head is replaced with an electronic meter that directly gives the reading.



27-4 Resistance-Capacitance (RC) Circuits

Charging a capacitor:

$$\mathcal{E} - iR - q/C = 0$$

$$i = dq/dt \Rightarrow$$

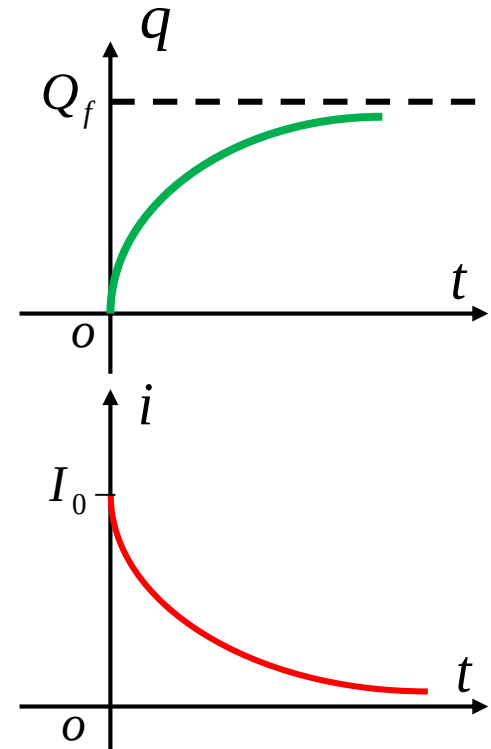
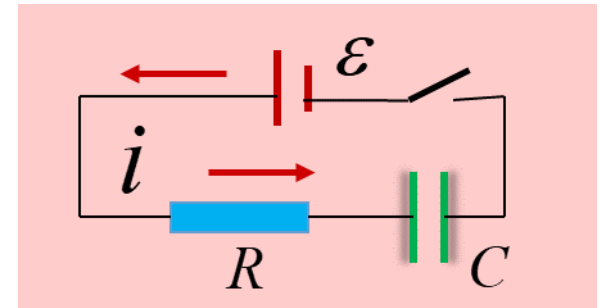
$$q = Q_f (1 - e^{-t/\tau})$$
$$i = I_0 e^{-t/\tau}$$

$$\tau = RC$$

time constant/relaxation time



Half of the energy supplied by the power source is stored in the capacitor, and the other half is consumed by the resistor.



27-4 Resistance-Capacitance (RC) Circuits

Charging a capacitor:

$$\varepsilon - iR - q/C = 0, \quad i = dq/dt$$

$$\varepsilon - R dq/dt - q/C = 0$$

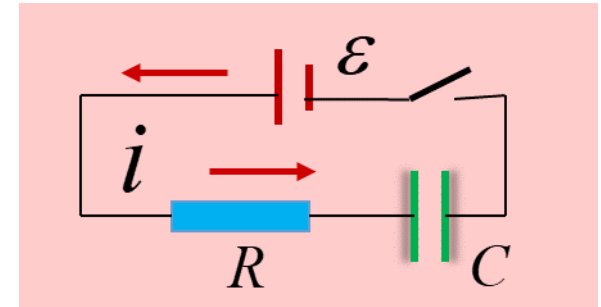
$$dq/dt = -(q - C\varepsilon)/RC, \quad dq/(q - C\varepsilon) = -dt/RC$$

$$\int_0^q \frac{dq}{q - C\varepsilon} = -\int_0^t \frac{dt}{RC}$$

$$\ln \frac{q - C\varepsilon}{-C\varepsilon} = -\frac{t}{RC}, \quad q = C\varepsilon(1 - e^{-t/RC})$$

$$i = dq/dt = \frac{\varepsilon}{R} e^{-t/RC}, \quad \tau = RC$$

$$q = Q_f(1 - e^{-t/\tau}), \quad i = I_0 e^{-t/\tau}$$



$$\left[\begin{array}{l} \varepsilon = \int_{-}^{+} \vec{E}_n \cdot d\vec{l} \\ W_e = Q_f \varepsilon \\ U = \frac{1}{2} Q_f \varepsilon \end{array} \right]$$

Half of the energy supplied by the power source is stored in the capacitor, and the other half is consumed by the resistor.



Example-4

4. A resistor with resistance $10\text{-M}\Omega$ is connected in series with a capacitor with capacitance $1.0\text{-}\mu\text{F}$ and a battery with emf 12.0-V as shown in Fig. Before the switch is closed at time $t = 0$, the capacitor is uncharged. (a) What is the time constant? (b) What fraction of the final charge is on the plates at time $t = 46\text{s}$? (c) What fraction of the initial current remains at $t = 46\text{s}$.

Solution:

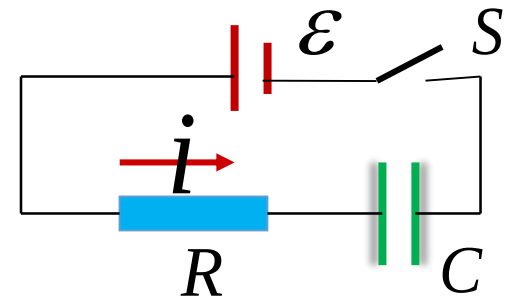
$$a) \tau = RC = 10 \times 10^6 \times 1.0 \times 10^{-6} = 10\text{s}$$

$$b) t = 46\text{s} = 4.6\tau, \quad q = Q_f (1 - e^{-t/\tau})$$

$$q / Q_f = 1 - e^{-t/\tau} = 1 - e^{-4.6} = 99\%$$

$$c) t = 46\text{s} = 4.6\tau, \quad i = I_0 e^{-t/\tau}$$

$$i / I_0 = e^{-t/\tau} = e^{-4.6} = 1.0\%$$



$$q = Q_f (1 - e^{-t/\tau}), \quad i = I_0 e^{-t/\tau}$$



27-4 Resistance-Capacitance (RC) Circuits

Discharging a capacitor:

$$0 - iR - q/C = 0, \quad i = dq/dt$$

$$Rdq/dt + q/C = 0$$

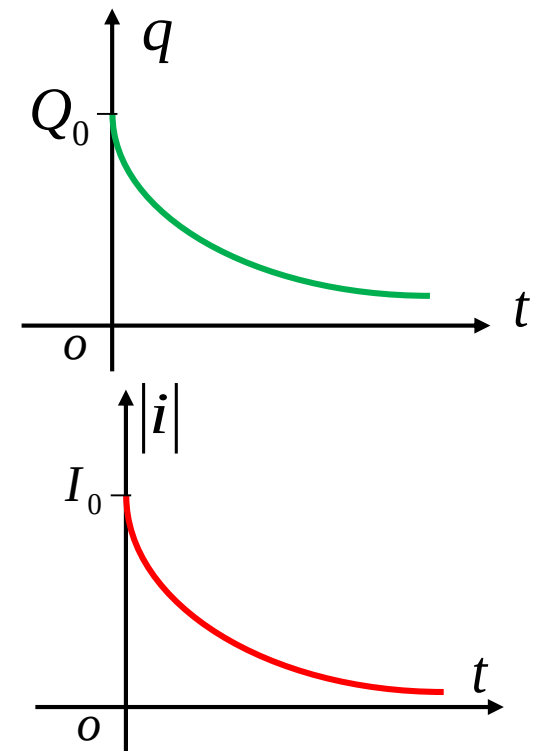
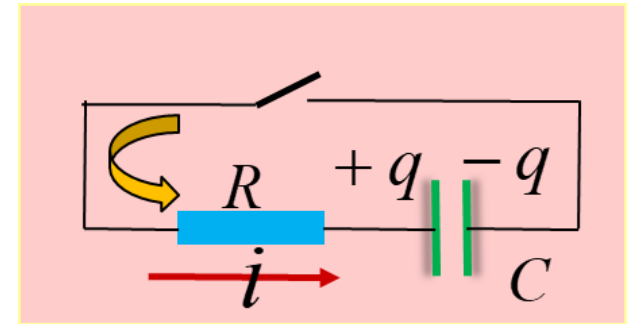
$$dq/dt = -q/RC, \quad dq/q = -dt/RC$$

$$\int_{Q_0}^q \frac{dq}{q} = -\int_0^t \frac{dt}{RC}$$

$$\ln \frac{q}{Q_0} = -\frac{t}{RC}, \quad q = Q_0 e^{-t/RC}$$

$$i = dq/dt = -\frac{Q_0}{RC} e^{-t/RC}, \quad \tau = RC$$

$$q = Q_0 e^{-t/\tau}, \quad i = -I_0 e^{-t/\tau}$$



Example-5

5. The resistor and capacitor in Example 5 are reconnected as shown in Figure. The capacitor is originally given a charge of $5.0\text{-}\mu\text{C}$ and then discharged by closing the switch at $t = 0$. (a) At what time will the charge be equal to $0.50\text{ }\mu\text{C}$? (b) What is the current at this time ?

Solution:

$$a) R = 10\text{M}\Omega, C = 1.0\text{ }\mu\text{F}, \tau = RC = 10\text{s}$$

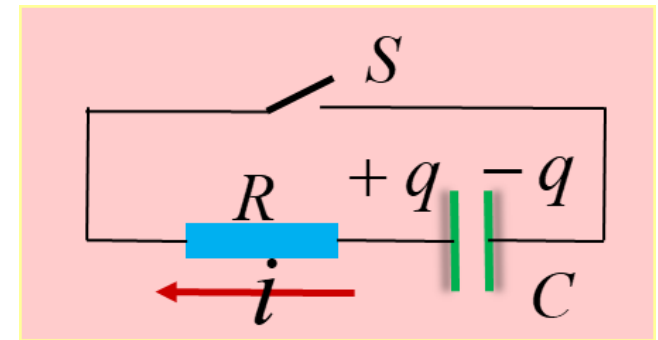
$$Q_0 = 5.0\text{ }\mu\text{C}, q = 0.5\text{ }\mu\text{C}$$

$$q = Q_0 e^{-t/\tau}, \quad e^{-t/\tau} = q / Q_0 = 0.10$$

$$t = -\tau \ln 0.10, \quad t = -10 \ln 0.10 = 23\text{s}$$

$$b) i = I_0 e^{-t/\tau}, \quad I_0 = \frac{Q_0}{RC} = \frac{5.0 \times 10^{-6}}{10} = 5.0 \times 10^{-7} \text{ A}$$

$$i = I_0 e^{-t/\tau} = 5.0 \times 10^{-7} \times 0.10 = 5.0 \times 10^{-8} \text{ A}$$



$$q = Q_0 e^{-t/\tau}, \quad i = -I_0 e^{-t/\tau}$$



